

Ahmed Abdelrahman

Cairo, Egypt (UTC+3) | +201024180920 | ahmedabdelrhman232@gmail.com |
<https://www.linkedin.com/in/ahmed-abdelrahman-89bbb3203/> | <https://kennycandra.dev>

Skills

languages

JavaScript, TypeScript, Go

frontend

React, Next.js, Tailwind CSS

backend

Node.js, Express.js, REST APIs,
Event-Driven Architecture

databases

PostgreSQL, Redis

architecture

DDD, Hexagonal Architecture,
Distributed Systems, Microser-
vices

infrastructure

AWS (ECS Fargate, ECR, RDS,
ElastiCache, Lambda, SSM),
Docker, GitHub Actions CI/CD

methodologies

Agile, Product-Oriented Think-
ing, AI-Driven Engineering,
CI/CD, Automated Testing

Education

Self-Taught Software Engineer

Work Experience

Vogel Edits

2025-01 - Present

Full-Stack Engineer (Adobe Extensions)

- Built and shipped production Adobe After Effects & Premiere Pro extensions, including Vanish BG — a real-time AI background removal tool running ML inference inside a sandboxed CEP environment; published on aescripts.com.
- Designed a Node.js sidecar process architecture that circumvents Adobe's restricted CEP runtime by handling AI inference externally, using local HTTP for IPC and streaming results back to the React UI in real time.
- Built and submitted a TTS Premiere Pro extension using Node.js with local ONNX model inference, designing the full audio pipeline from text input to rendered timeline clip.
- Collaborated asynchronously with designers and stakeholders in an agile workflow, shipping production-ready tools end-to-end.

Freelance

2024-11 - 2025-01

Full-Stack Engineer

- Built a full-stack tourism booking platform using Next.js, TypeScript, and Node.js with TypeORM for type-safe PostgreSQL access — delivered autonomously from conception to deployment.
- Applied PostgreSQL indexing strategies and EXPLAIN ANALYZE to achieve sub-100ms API response times under concurrent load.
- Integrated Paymob payment gateway with idempotent transaction handling and webhook verification, reducing checkout friction by 40%.
- Improved SEO and Core Web Vitals to 95+ Lighthouse score using Next.js SSR, image optimization, and lazy loading.

Projects

RootCluster — AI-Powered Log Analysis Platform <https://rootcluster.dev>

Distributed log ingestion and root-cause analysis platform processing 1M+ events/hour with sub-second latency.

Tech Stack: TypeScript, Node.js, PostgreSQL, Redis, AWS ECS Fargate, Docker, BullMQ

- Designed and deployed event-driven microservices pipeline on AWS ECS Fargate: normalize → embed → cluster → merge → root cause, with retries, exponential backoff, DLQs, and idempotent processing.
- Reduced ONNX embedding time from 20+ minutes to under 1 minute via batch processing, Redis caching, and worker autoscaling.
- Built full CI/CD pipeline with GitHub Actions OIDC, ECR, SSM Parameter Store, RDS PostgreSQL, and ElastiCache.
- Applied CAP theorem tradeoffs — PostgreSQL for strong consistency, Redis for high-availability caching — and PostgreSQL internals (MVCC, isolation levels, indexing) for a high-throughput write pipeline.

Medical CRM — AI-Enhanced Prescription Platform

Full-stack CRM handling diagnoses, prescriptions, and patient records with drug interaction validation across 500+ medications.

Tech Stack: React, Node.js, PostgreSQL, JWT, RBAC

- Built full-stack SPA with React and Node.js implementing JWT auth with access/refresh token rotation and RBAC across three permission levels.
- Designed graph-based drug interaction lookup in PostgreSQL for sub-100ms validation across 500+ medications.
- Optimized dashboard queries for 10K+ monthly records using composite indexes and window functions.